Railway Analytics Project

Database Design & Technical Documentation

Database Management System: MySQL
Hosting Provider: FreeSQLDatabase.com
Database Name: sql12839745
Server: sql12.freesqldatabase.com
Port: 3306
Storage Engine: InnoDB
Character Set: utf8mb4

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1. Database Overview

The Railway Analytics project uses a relational MySQL database designed with a dimensional modeling structure.

The database consists of three dimension tables and one central fact table. The dimension tables store descriptive information, while the fact table stores individual ticket purchase and journey records.

The database serves as the source for Power BI data modeling, DAX calculations, and interactive reporting.

2. Database Architecture

Table	Type	Purpose
Dim_Date	Dimension	Stores calendar attributes for time-based analysis.
Dim_Route	Dimension	Stores departure stations and arrival destinations.
Dim_Ticket	Dimension	Stores ticket, railcard, and ticket-class information.
Fact_Journey	Fact	Stores ticket purchase and journey transactions.

Logical Relationship Structure

The fact table is related to the three dimension tables through their corresponding key columns.

Fact Table Column	Dimension Table	Dimension Key

Ticket_ID	Dim_Ticket	Ticket_ID
Route_Key	Dim_Route	Route_Key
Date_of_Purchase	Dim_Date	Date
Date_of_Journey	Dim_Date	Date

Dim_Date serves two analytical roles: purchase date and journey date.

Relationship implementation: These relationships are defined in Power BI. Foreign key constraints are not physically enforced in MySQL.

3. Database Tables and Record Counts

The following record counts were verified using SQL COUNT(*) queries against the online database.

Table	**Records**
Dim_Date	145
Dim_Route	64
Dim_Ticket	24
Fact_Journey	31,653
Total	**31,886**

4. Table Definitions

4.1 Dim_Date

Purpose: Stores calendar information used for purchase-date and journey-date analysis.

Column	**Data Type**	**Constraint**
Date	DATE	Primary Key, NOT NULL
Year	SMALLINT	NOT NULL
Quarter	TINYINT	NOT NULL
Month	TINYINT	NOT NULL

Month_Name	VARCHAR(10)	NOT NULL
Day	TINYINT	NOT NULL
Day_Name	VARCHAR(10)	NOT NULL

Primary Key: Date

Record Count: 145

4.2 Dim_Route

Purpose: Stores railway route information, including departure stations and arrival destinations.

Column	Data Type	Constraint
Route_Key	VARCHAR(60)	Primary Key, NOT NULL
Departure_Station	VARCHAR(40)	NOT NULL
Arrival_Destination	VARCHAR(40)	NOT NULL

Primary Key: Route_Key

Record Count: 64

4.3 Dim_Ticket

Purpose: Stores ticket-related attributes used to analyze ticket categories and purchasing characteristics.

Column	Data Type	Constraint
Ticket_ID	VARCHAR(8)	Primary Key, NOT NULL
Railcard	VARCHAR(12)	NOT NULL
Ticket_Class	VARCHAR(12)	NOT NULL
Ticket_Type	VARCHAR(10)	NOT NULL

Primary Key: Ticket_ID

Record Count: 24

4.4 Fact_Journey

Purpose: Stores individual ticket purchase transactions and their associated journey details. Each row represents one ticket purchase.

Column	Data Type	Constraint
Transaction_ID	VARCHAR(23)	Primary Key, NOT NULL
Ticket_ID	VARCHAR(8)	NOT NULL
Date_of_Purchase	DATE	NOT NULL
Purchase_Type	ENUM('Online','Station')	NOT NULL
Payment_Method	ENUM('Contactless','Credit Card','Debit Card')	NOT NULL
Time_of_Purchase	TIME	NOT NULL
Price	SMALLINT UNSIGNED	NOT NULL
Date_of_Journey	DATE	NOT NULL
Departure_Time	TIME	NOT NULL
Arrival_Time	TIME	NOT NULL
Refund_Request	ENUM('No','Yes')	NOT NULL
Actual_Arrival_Time	TIME	NOT NULL
Journey_Status	ENUM('On Time','Delayed','Cancelled')	NOT NULL
Reason_For_Delay	ENUM('No Delay','Weather','Signal Failure','Staff Shortage','Technical Issue','Traffic')	NOT NULL
Route_Key	VARCHAR(60)	NOT NULL
Actual_Arrival_Status	ENUM('Reported','Cancelled')	NOT NULL

Primary Key: Transaction_ID

Record Count: 31,653

5. Primary and Foreign Key Design

Primary Keys

Each table has a primary key that uniquely identifies its records.

Table	Primary Key
Dim_Date	Date
Dim_Route	Route_Key
Dim_Ticket	Ticket_ID
Fact_Journey	Transaction_ID

Foreign Key Relationships

The following logical relationships connect the fact table to the dimension tables:

Fact Table Foreign Key Column	Referenced Table	Referenced Column
Ticket_ID	Dim_Ticket	Ticket_ID
Route_Key	Dim_Route	Route_Key
Date_of_Purchase	Dim_Date	Date
Date_of_Journey	Dim_Date	Date

Implementation note: These columns are designed to establish foreign key relationships, but the corresponding MySQL foreign key constraints are not enabled. This was an intentional design decision to reduce additional index storage under the hosting provider's limited database storage allowance. Referential integrity should therefore be checked using validation queries.

6. Data Loading

The source data was prepared in Excel and exported into CSV files.

The tables were created in the online MySQL database, and the CSV files were imported using MySQL LOAD DATA LOCAL INFILE through the MySQL command-line client.

The imported data was verified using record-count queries.

7. SQL Validation Queries

7.1 Verify Database and Tables

```
SELECT DATABASE();
```

```
SHOW TABLES;
```

7.2 Verify Record Counts

```
SELECT 'Dim_Date' AS Table_Name, COUNT(*) AS Total_Rows
FROM Dim_Date

UNION ALL

SELECT 'Dim_Route', COUNT(*) FROM Dim_Route

UNION ALL

SELECT 'Dim_Ticket', COUNT(*) FROM Dim_Ticket

UNION ALL

SELECT 'Fact_Journey', COUNT(*) FROM Fact_Journey;
```

7.3 Inspect Table Definitions

```
SHOW CREATE TABLE Dim_Date;
SHOW CREATE TABLE Dim_Route;
SHOW CREATE TABLE Dim_Ticket;
SHOW CREATE TABLE Fact_Journey;
```

7.4 Check Duplicate Primary Keys

```
SELECT `Date`, COUNT(*) AS Duplicate_Count
FROM Dim_Date
GROUP BY `Date`
```

```
HAVING COUNT(*) > 1;


SELECT Route_Key, COUNT(*) AS Duplicate_Count
FROM Dim_Route
GROUP BY Route_Key
HAVING COUNT(*) > 1;


SELECT Ticket_ID, COUNT(*) AS Duplicate_Count
FROM Dim_Ticket
GROUP BY Ticket_ID
HAVING COUNT(*) > 1;


SELECT Transaction_ID, COUNT(*) AS Duplicate_Count
FROM Fact_Journey
GROUP BY Transaction_ID
HAVING COUNT(*) > 1;
```

Expected result: No duplicate records.

7.5 Validate Referential Integrity

The following queries identify fact records whose foreign key values do not exist in their corresponding dimension tables.

```
-- Invalid Ticket References
SELECT COUNT(*) AS Invalid_Ticket_References
FROM Fact_Journey f
LEFT JOIN Dim_Ticket t
    ON f.Ticket_ID = t.Ticket_ID
WHERE t.Ticket_ID IS NULL;
```

```
-- Invalid Route References
SELECT COUNT(*) AS Invalid_Route_References
FROM Fact_Journey f
LEFT JOIN Dim_Route r
    ON f.Route_Key = r.Route_Key
WHERE r.Route_Key IS NULL;

-- Invalid Purchase Date References
SELECT COUNT(*) AS Invalid_Purchase_Date_References
FROM Fact_Journey f
LEFT JOIN Dim_Date d
    ON f.Date_of_Purchase = d.`Date`
WHERE d.`Date` IS NULL;

-- Invalid Journey Date References
SELECT COUNT(*) AS Invalid_Journey_Date_References
FROM Fact_Journey f
LEFT JOIN Dim_Date d
    ON f.Date_of_Journey = d.`Date`
WHERE d.`Date` IS NULL;
```

Expected result: Zero invalid references for each relationship.

7.6 Inspect Sample Records

```
SELECT * FROM Dim_Date LIMIT 5;
SELECT * FROM Dim_Route LIMIT 5;

SELECT * FROM Dim_Ticket LIMIT 5;
```

```
SELECT * FROM Fact_Journey LIMIT 5;
```

8. Power BI Integration

Power BI Desktop connects to the online MySQL database using the MySQL Database connector.

Connection configuration:

Setting	Value
Server	sql12.freesqldatabase.com:3306
Database	sql12839745
Authentication	Database
Username	sql12839745

The four tables are imported into Power BI, where the relationships are configured in the data model.

Date Relationship Consideration

Because Fact_Journey contains two date columns connected to the same Dim_Date table, the model must account for the two date roles.

The purchase-date relationship and journey-date relationship cannot both be active between the same two tables in the standard Power BI relationship model.

One relationship can be active, while the other is inactive and activated in relevant DAX measures using USERELATIONSHIP(). Alternatively, two separate role-playing date tables can be created.

9. Database Design Summary

Design Component	Implementation
Database Platform	MySQL
Storage Engine	InnoDB
Database Tables	4
Dimension Tables	3

Fact Tables	1
Total Records	31,886
Primary Keys	4
Foreign Key Constraints in MySQL	Not enabled
Logical Relationships	4
Data Import Format	CSV
Data Loading Method	LOAD DATA LOCAL INFILE
BI Integration	Power BI Desktop

10. Conclusion

The Railway Analytics database implements a dimensional relational structure consisting of three dimension tables and one central fact table.

With 31,886 records, including 31,653 journey transactions, the database provides the foundation for analyzing railway purchases, routes, ticket categories, and journey performance.

Primary keys enforce record uniqueness, while logical relationships connect the fact table to the dimension tables through Power BI. SQL validation queries support record-count verification, duplicate detection, and referential integrity checks.

The database is integrated with Power BI to support data modeling, analytical calculations, and interactive business intelligence reporting.